

Luka Matijević

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Current Employment

2023 – now **Teaching Assistant**, University of Zagreb, Faculty of Science, Department of Physics, Zagreb, Croatia.

Research Interests

◦ Extragalactic astrophysics

Galaxy-environment interplay in the local Universe,
Star formation rates / histories of nearby galaxies,
ISM/IGM properties,
Photometry and spectroscopy of galaxies

Education

2023 – now **Doctoral study of Physics - Astrophysics**, University of Zagreb, Faculty of Science, Department of Physics, Zagreb, Croatia.

Doctoral thesis title: *"Environmental effects on the physical and chemical properties of NGC 2276 on small spatial scales"*

2017 – 2023 **Integral Undergraduate and Graduate Study of Physics**, University of Zagreb, Faculty of Science, Department of Physics, Zagreb, Croatia.

Master's thesis title: *"Masses of supermassive black holes in quasars"*

Teaching Experiences

2023 – now **Teaching Assistant**, University of Zagreb, Faculty of Science, Department of Physics, Zagreb, Croatia.

- subjects: Physics 1 (Chemistry Department), Physics 2 (Mathematics Department), Introduction to Astrophysics, Stellar Astrophysics, Introductory Physics Lab 1

2018 – 2019 **Junior Teaching Assistant**, University of Zagreb, Faculty of Science, Department of Physics, Zagreb, Croatia.

- Assisting younger students in "Computing Laboratory" with programming tasks in C and Python

Research & Employment Experiences

2023 – now **Faculty of Science - Department of Physics, Zagreb, Croatia**, *PhD research*, under the supervision of Asst. Prof. Neven Tomičić and Dr. Antonino Marasco.

- studying the influence of large-scale environmental effects on the star formation in galaxy NGC 2276 and NGC 2300 group

2021 **Ruđer Bošković Institute, Zagreb**, *Student Autumn Projects*, under supervision of Vibor Jelić, several months project.

- Studying magnetohydrodynamical simulation data of interstellar medium

- Comparison of simulation results with observational telescope data on different frequencies

- 2021 **Xylon d.o.o., Zagreb**, *Student Internship*, under supervision of Josip Šanko, several-month project.
 - Creating a MATLAB-Blender interface for animations using simulated camera data on vehicles.
- 2019 **Summer School of Science S3++**, **Požega**, *Building Your Own Plasma Speaker*, as workshop leader, one-day project.
 - Teaching Summer School students how a speaker works and building a Plasma Speaker with them using microelectronic components
- 2017 **Summer School of Science S3++**, **Požega**, *Investigation of aerodynamic properties*, under supervision of Krunoslav Šešet, seven-day project.
 - Learned about aerodynamics and how various car components affect aerodynamic properties
 - 3d printed two cars with different aerodynamic properties and used a wind tunnel with liquid nitrogen to test them

Successful Observing Proposals

- Asiago **PI of Asiago Cima-Ekar AFOSC proposal**, 45h.
 title: *"Asiago Spectroscopic Investigation for Analysis of Group Objects (ASIAGO)"*
- VLA **PI of L-band, D-configuration proposal**, ID: VLA/26A-162, 12.5h.
 title: *"Mapping the neutral gas properties in the longest ram pressure stripped tail"*
- WHT **Main Co-PI of LIFU/WEAVE proposal**, PI: Scott Trager, ID: WS2025B2-012, 22h.
 title: *"The Difficult Life Of Galaxies In Groups: The Peculiar Case Of NGC 2276"*

Honors & Awards

- Project PMF-YOUTHCONF-2026-03, NextGenerationEU Institutional grant of the University of Zagreb Faculty of Science, financing a poster presentation at the Olympian Symposium 2026, Greece, May 2026
- Project HRZZ-MOBDOK-2023-8006, Mobility project for doctoral students awarded by Croatian Science Foundation, working at the Astronomical Observatory in Padova, Italy, February-July 2025
- Municipality of Velika scholarship for students in middle and higher education, Croatia, academic year 2022/2023, 2021/2022, 2018/2019
- Student DIGI Awards as a member of Tongueo project team, 2022
- STEM national scholarship for students in higher education, Croatia, academic year 2019/2020, 2017/2018

Activities & Memberships

- Observations with an optical 1.2m telescope in long-slit configuration, Asiago, Italy, 2026
- Muzza Week of Science, Zagreb, Croatia, 2026
- Member of Croatian Astronomical Society (HAD), Zagreb, Croatia, 2025 -
- Member of Tongueo project developed inside Spock Incubator on the Faculty of Electrical Engineering and Computing, Zagreb, Croatia, 2022
- Brainhack hackathon, Zagreb, Croatia, 2021
- Technical Assistant, Summer School of Science S3++, Požega, Croatia, 2019
- Member of Society for Out of Frame Education (Društvo za edukaciju van okvira), Zagreb, Croatia, 2018 - 2019
- Volunteering at Summer School of Science S3++, Požega, Croatia, 2018

Finished Courses

- 2022 **The Diversity of Exoplanets**, Online course on Coursera organised by the University of Geneva.
- Gained knowledge about exoplanet formation, various methods of discovering exoplanets, exoplanetary atmospheres, etc.
- 2022 **Data-driven Astronomy**, Online course on Coursera organised by The University of Sydney.
- Gained experience in using large datasets, creating SQL queries and machine learning methods
- Worked with modern astronomical data such as AT20G Bright Source Sample (BSS) catalogue, SuperCOSMOS all-sky galaxy catalogue, NASA Exoplanet database and Sloan Digital Sky Survey data.
- 2022 **Illuminative**, Mobile app development course organised by EESTEC LC Zagreb.
- Gained knowledge about creating different types of applications in Kotlin and Java programming languages.
- 2020 **Elements of AI**, Online course organised by University of Zagreb and University of Helsinki.
- Learned about various techniques of which AI is comprised such as neural networks, machine learning, etc.
- 2020 **3D modelling course in Autodesk 3ds Max**, Online course organised by Studium College.
- Learned to use Autodesk 3ds Max software in 3d modelling, UV unwrapping, sculpting, retopology, texturing, lighting, animation and rendering

Skills

- **Programming**
Usage of Python, C and Matlab programming languages
- **3d modelling**
Usage of Blender and 3ds Max software in modelling, UV unwrapping, texturing and rendering as well as beginner animation
- **Other tools:**
MS Office, L^AT_EX, Linux/UNIX, QtiPlot, SAOImageDS9, PyQSOFit, QFitsView, ^{3D}Barolo, Galfit, Bagpipes, SExtractor, PyLongslit, Pyphot, numpy, scipy, pandas, astropy, matplotlib, aplpy, Topcat

Languages

- **Mother tongue:** Croatian
- **Foreign Language #1:** English
(Reading: B2, Listening: B2, Speaking: B2, Writing: B2)
- **Foreign Language #2:** Japanese
(Reading: A1.1, Listening: A1.1, Speaking: A1.1, Writing: A1.1)

Other

- European driving licence for AM, B, F, G vehicle classes

First Author Publications

- 2026/08 **Matijević, L.**, Marasco, A., Tomičić, N. et al., arXiv:2608.22951, accepted for publication in A&A in August 2026 .
"The star formation history of NGC 2276: Comparison between SED modeling and hydrodynamical simulations"
- 2026/02 **Matijević, L.**, Tomičić, N., Marasco, A. et al., Astronomy & Astrophysics, 707 (2026), A40-A52.
"The competing influences of ram pressure and tidal interaction in NGC 2276"

Other Publications

- Lassen, A. E.; Vulcani, B., et al. **incl. Matijević, L.**, Astronomy & Astrophysics, 710, (2026), A215-A235, *"With arms wide open: A VLT/MUSE view of the mechanisms driving unwinding spiral arms in cluster galaxies"*
- Lassen, A. E.; Smith, R., et al. **incl. Matijević, L.**, Astronomy & Astrophysics, 706 (2026), A85-A100, *"Distinguishing ram pressure from gravitational interactions: Applying the size-shape difference method to real galaxies"*
- Roberts, I. D.; van Weeren, R. J., et al. **incl. Matijević, L.**, Astronomy & Astrophysics, 689 (2024), A22-A35, *"A 100 kpc ram pressure tail trailing the group galaxy NGC 2276"*

University Diplomas

- Master's Thesis, *"Masses of supermassive black holes in quasars"*, Link to repository: [click here](#)

Conferences

- Poster presentation, "*The influence of the environment on the morphology and ISM properties of the lopsided galaxy NGC 2276*", The Olympian Symposium, Paralia Katerini, Greece, May 2026
- Poster presentation, "*The influence of the environment on the lopsided galaxy NGC 2276*", Multi-phase, Multi-temperature and Complex, Olbia, Italy, October 2025
- Poster presentation, "*The influence of the environment on the lopsided galaxy NGC 2276*", CL2025: Entering a Golden Age of Galaxy Cluster Studies, Taipei, Taiwan, September 2025
- Oral presentation, "*The effects of ram pressure and tidal forces on the lopsided galaxy NGC 2276*", Croatian Astronomical Society Scientific Meeting, Rijeka, Croatia, November 2024
- Poster presentation, "*NGC 2276 through the eyes of the Hubble Space Telescope: The effects of ram pressure and tidal forces on the various stellar and gas tracers*", GASPIsa 2024 scientific conference, Pisa, Italy, May 2024
- Oral presentation, "*The peculiar lopsided galaxy NGC 2276*", PhD Student Symposium, Zagreb, Croatia, April 2024

Invited Lectures

- Seminar at the INAF-OAPd, Padua, Italy, 2026
- Extragalactic meeting at the Arcetri Observatory, Florence, Italy, 2024
- "*Active galaxies and black holes*", Outreach & public presentation, Observatory Zagreb, Zagreb, Croatia, 2024
- "*Active galaxies and black holes*", Outreach & public presentation, Požega Gymnasium, Croatia, 2023

Reference contacts

- **Asst. Prof. Neven Tomičić**
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- **Dr. Antonino Marasco**
INAF Istituto Nazionale di Astrofisica Osservatorio Astronomico di Padova, Padova, Italy
PhD advisor
`antonino.marasco@inaf.it`
- **Prof. Željko Ivezić**
University of Washington, Dept. of Astronomy, Box 351580, Seattle, WA 98195, USA
Supervisor for a Rubin LSST project
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